

Assignment V

Machine Learning

February 21, 2026

Abstract

These exercises aim to review some unsupervised learning models. Please upload your solutions to Classroom before March 3rd.

Problem 1

Read [1] about feature selection family methods. Then, write down your personal criteria to choose one over the other.

Problem 2

Consider the following function:

$$f(x, y) = \frac{RMSE(x, y)}{\|x\|_2}, \quad x, y \in \mathbb{R}^n, \|x\|_2 \neq 0.$$

Write down the domain and image of f . What is a “good” value/range for f so we can say x is very similar to y ?

Problem 3

Consider the divorce forecasting dataset [2]. Next:

- a) Solve the classification problem
- b) Identify the most relevant features for the classification. The goal here is to reduce the number of features employed while maintaining high classification accuracy.

Write down all your assumptions and discuss your findings.

Problem 4

Repeat problem 3, but this time create new features. The goal is the same as before: reduce the number of features used while maintaining high classification accuracy. Was it possible to decrease the number of features using this approach? Write down all your assumptions and discuss your findings.

Problems 5 - 6

Repeat problems 3 and 4 considering the car price forecasting problem [3].

Problem 7

Use the K-Means and DBSCAN algorithms to make image compression by colors, as explained in [4]. Write down all your assumptions, make comparisons, and discuss your findings.

Problem 8

Is K-Means too sensitive to outliers because of the mean operator? Let's find out:

- a) Build a dataset with some outliers showing how sensible K-Means is.
- b) Code a K-Median clustering algorithm, which is basically the K-Means but it uses the median instead of the mean.
- c) Apply your K-Median to your data. Does it cluster any better?

Write down all your assumptions and discuss your findings.

Problem 9

Consider the problem of gender prediction [5]. Do the following

1. Make a clustering process over the features considering two clusters. Do your results align with the known labels?
2. Try to identify all the outliers.
3. Select some random data and flip their label. Then, try to identify all the outliers. Was your method able to successfully identify the flipped data?

Write down all your assumptions, make comparisons, and discuss your findings.

References

- [1] https://medium.com/@learnwithwhiteboard_digest/filter-vs-wrapper-vs-embedded-methods-for-feature-selection-8cc21e2174f7
- [2] <https://www.kaggle.com/datasets/andrewmvd/divorce-prediction>
- [3] <https://www.kaggle.com/datasets/hellbuoy/car-price-prediction>
- [4] <https://medium.com/codex/image-compression-with-k-means-clustering-48e989055729>
- [5] <https://www.kaggle.com/datasets/krishnaraj30/weight-and-height-data-outlier-detection>